

ESE 5760 Final Project Proposal

Trend-Based Comparison of Emerging Memory Technologies in DESTINY: Scaling, 2D/3D Stacking, and TSV-Sensitive Tradeoffs

Tiebing Tang

Yishu Wang

Zhengyu Liu

April 20, 2026

1 Project Proposal

This project studies how different memory technologies and architectural assumptions affect cache-level performance, energy, and area using the DESTINY simulator as a parametric evaluation platform. Instead of centering the project on large-scale debugging of the simulator codebase, we focus on extracting and comparing quantitative trends from controlled input variations. This direction aligns with the final project requirement that emphasizes original analysis and quantitative evaluation rather than a pure literature survey.

Our study will focus on a subset of memory technologies supported and practically evaluable in DESTINY, including eDRAM and selected non-volatile memories such as STT-RAM, ReRAM/RRAM, and PCRAM where model support is available. Starting from a validated baseline cache configuration, we will perform three main groups of experiments. First, we will compare different memory technologies under the same or comparable cache capacity to identify tradeoffs in latency, dynamic energy, leakage power, and area. Second, we will apply representative process-node scaling assumptions to study how relative trends evolve across technology generations. Third, we will compare 2D and 3D organizations, including 2-layer and 4-layer cases, to evaluate when vertical integration provides measurable benefit.

A key extension of the project is the inclusion of TSV-sensitive analysis in the 3D study. Since the projected benefit of 3D memory strongly depends on vertical interconnect assumptions, we will examine whether changes in TSV-related parameters materially alter the simulator’s conclusions. This allows us not only to compare 2D and 3D organizations, but also to assess how robust DESTINY’s 3D predictions are with respect to its interconnect assumptions.

Where possible, we will use representative literature data and prototype trends as sanity checks, especially for parameters such as cell area, retention-related assumptions, and scaling direction. However, the main contribution of the project is not to reproduce one exact hardware implementation, but to identify robust comparative trends and determine which conclusions are stable versus

highly assumption-dependent. The final outcome will be a quantitatively grounded comparison of memory technologies and stacking strategies, along with a discussion of what DESTINY can and cannot reliably predict under these scenarios.

2 Research Questions

1. How do different memory technologies supported by DESTINY compare under the same cache capacity and comparable architectural assumptions?
2. How do representative process-node scaling assumptions change the relative tradeoffs among these memory technologies?
3. How do 3D stacking assumptions, especially layer count and TSV-related overhead, affect the projected benefits of different memory organizations?

3 Proposed Report Outline

1. **Abstract**
2. **Introduction and Motivation**
3. **Background on Memory Technologies and DESTINY Modeling Assumptions**
4. **Methodology and Baseline Configuration**
5. **Results I: Cross-Technology Comparison at Fixed Capacity**
6. **Results II: Scaling Trends Across Representative Process Nodes**
7. **Results III: 2D/3D Comparison and TSV-Sensitive Analysis**
8. **Discussion: Robust Trends vs. Assumption-Dependent Conclusions**
9. **Limitations and Possible Model Gaps**
10. **Conclusion**

4 Poster Plan

Given the April 26 poster deadline, the poster will emphasize the most stable and interpretable results that can be communicated clearly in a visual format. The poster is expected to contain four main parts:

1. **Motivation and Scope:** why emerging memory comparison matters and why DESTINY is used as the evaluation platform.

2. **Methodology:** baseline configuration, parameter-variation workflow, and output metrics such as latency, dynamic energy, leakage power, and area.
3. **Core Results:**
 - fixed-capacity cross-technology comparison,
 - representative node-scaling trends,
 - 2D vs. 3D and TSV-sensitive comparisons.
4. **Early Takeaways:** a small set of concise conclusions highlighting which trends appear robust and which are highly assumption-dependent.

The poster will prioritize 3–5 strong figures over exhaustive coverage. Additional sensitivity studies, deeper literature grounding, and broader discussion of limitations will be reserved for the final writeup.

5 Updated Timeline

Because the project has already moved beyond initial exploration, the remaining schedule is organized around the poster and final report deadlines.

April 20 – April 22

- Finalize the narrowed project scope and baseline experiment matrix.
- Confirm which memory technologies and process nodes can be evaluated reliably in DESTINY.
- Select the result types that are realistic to complete before the poster deadline.

April 22 – April 24

- Run the main cross-technology comparison at fixed capacity.
- Run the first set of node-scaling experiments.
- Run the core 2D/3D experiments with preliminary TSV-sensitive sweeps.
- Start generating publication-quality plots and tables.

April 24 – April 26

- Consolidate the most stable and interpretable results.
- Draft poster sections: motivation, methodology, key results, and early conclusions.
- Reduce content to a poster-friendly scope with 3–5 main figures.
- Finalize and submit the poster on April 26.

April 27 – April 30

- Extend experiments beyond the poster scope where needed.
- Add literature-based sanity checks and deeper discussion.
- Refine analysis of assumption-sensitive trends, especially TSV-related results.

May 1 – May 3

- Write and revise the final report.
- Integrate poster results into the final IEEE-style writeup.
- Add limitations, discussion, and final conclusions.
- Proofread and finalize the submission.

A Gantt Chart

The following Gantt chart shows the revised project plan from April 20 to May 3, centered around the poster deadline and the final writeup.

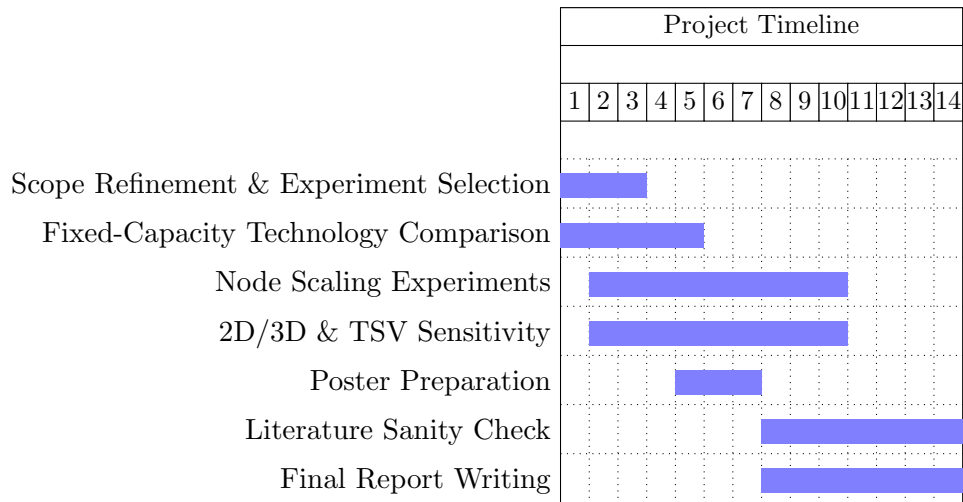


Figure 1: Revised project schedule based on the current phase of the project.